



Polymer brushes prepared by surface-initiated atom transfer radical polymerization of poly (N-isopropyl acrylamide) and their antifouling properties

M. Rahimi, M. Nasiri*

*Institute of Polymeric Materials, Sahand University of Technology, Tabriz, Iran
Faculty of Polymer Engineering, Sahand University of Technology, Tabriz, Iran*

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ABSTRACT

Poly (N-isopropyl acrylamide) (PNIPAAm) polymer as a thermo-responsive coating was synthesized by surface-initiated atom transfer radical polymerization (SI-ATRP) on glass substrates. The effects of chain grafting density and polymerization time on the fouling release behavior of a marine diatom were investigated. The PNIPAAm grafting and its thermo-responsiveness were confirmed by different analyses such as scanning electron microscopy (SEM), attenuated total reflectance Fourier transform infrared spectroscopy (ATR-FTIR), energy-dispersive X-ray spectroscopy, and water contact angle (WCA). Three-way ANOVA analysis reveals that all parameters, including polymerization time, UV irradiation time, and temperature are significant. The results show that by lowering the temperature below the lower critical solution temperature (LCST) of PNIPAAm, the microorganisms wholly detached from the surface. Therefore, PNIPAAm can release marine microorganisms, and it can be a suitable and environmentally friendly alternative for toxic antifouling coatings.

1. Introduction

Marine structures such as ships, oil platforms, and oceanographic systems are permanently exposed to fouling organisms. Biofouling has undesirable effects on these structures, so that reduces their performance. It raises the fuel consumption of ships and facilitates the potential transport of microorganisms to the regions that are not belonging to [1–3]. Antifouling coatings, to some extent, prevent the attachment of marine fouling to the surface of marine structures [4,5]. Most of these coatings use toxic materials that are harmful to the other microorganisms [6]. Due to the toxicity of antifouling paints, the use of these coatings has been prohibited by the International Marine Organization Convention; therefore, the use of environmentally friendly antifouling coatings is preferred. So far three general methods including biocides containing coatings, self-polishing, and biocide-free coatings have been used to prevent the marine fouling [7]. Currently, the most practical method is to use bactericides that are toxic substances and kill marine microorganisms. Recently, bactericidal synergists such as D-Amino acid, D-Methionine, and D-Tyrosine have been used to improve bactericidal performance [8]. While the bioactive materials extracted from marine microorganisms, especially from *Streptomyces*, show promising effect in inhibition of biofouling, small scale production

limits their commercial applications [9,10]. In self-polishing coatings, the biocides are released slowly by the degradation of the polymer; preventing the adhesion of microorganisms over a long period of time. Immobilization of the bioactive materials in the coatings is one of the strategies to improve their long-term activities. Due to the sensitivity of the bioactive material to the environmental conditions, the nature of the polymeric coating is critical. Different polymers such as hydrogels [11], latexes [12], and porous cellulose [13] were used for this purpose. Hydrogen peroxide as an environmentally friendly compound was used in self-polishing coatings, and while prevents some types of microorganisms, it is not useful for barnacles and tubeworms [14]. Nanotechnology is a unique tool that can control different properties of the coatings such as surface energy, conductivity, porosity, wettability, friction, and roughness. In comparison with soluble biocides in water, the nanoparticles are effectively immobilized in coatings, and the risk of their release to the environment is decreased [15]. Different nanoparticles, including ZnO [16], Ag [17], Cu [15], carbon nanotubes [16], TiO₂ [18–20], and SiO₂ [21], were used successfully in antifouling coatings. However, some of these nanoparticles are toxic. Among different nontoxic methods for producing antifouling coating, hydrophobic coatings are important ones [22]. The high roughness of the superhydrophobic coatings creates a layer of air acting as a barrier

* Corresponding author at: Sahand University of Technology, New Town of Sahand, Tabriz 5331817634, Iran.

E-mail address: m_nasiri@sut.ac.ir (M. Nasiri).

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between the coating and water. Therefore, the fouling process is inhibited to some extent [23,24]. Some chemical and physical features of hydrophobic coating designs were inspired by nature [25]. The antifouling properties of these coatings are limited to the features in the micron-scale size due to the technological limitations in nanometric scales [26]. Amphiphilic polymers containing zwitterionic groups, such as polyethylene glycol-based copolymers, have shown good antifouling properties [27]. Polypyrrole, as a conductive polymer, undergoes hydrophilicity change by electrical stimulation, preventing the adhesion of proteins and bacteria to the surface [28]. Despite the high demand for a nontoxic antifouling coating, until now, there is no efficient one [9].

Responsive polymers are an important group of polymers that undergo changes such as isomerization, chemical bond breakage, hydrogen bond, ionic strength and so on due to external stimuli such as light, temperature, electric current, magnetic field, pH, and chemical composition. Among these materials, thermo-responsive polymers especially PNIPAAm have been widely studied and used in various fields. PNIPAAm and its copolymers have been used in various fields of medicine such as drug delivery, cardiac tissue engineering [29,30], cell sheet in transplantation [31–35], smart injectable hydrogels and shape-memory polymers [36], thermo-responsive chromatography of proteins [37], cell separation [38]. They were also used in other high-technologies such as smart gating membranes [39] and Nanostructured surfaces using thermo-responsive nanogels [40]. Various types of PNIPAAm-based surfaces were prepared by different methods such as electron beam-induced polymerization, ATRP, RAFT polymerization or physical adsorption [41]. To the best of our knowledge, there is not any study in the application of PNIPAAm in the marine antifouling coatings. However, The effects of wettability, molecular weight and graft density of PNIPAAm brushes on protein adsorption were investigated [42].

The PNIPAAm based thermo-responsive polymers are well used in regenerative medicine, membranes, etc., and they exhibit good adhesion and release behavior against various human and animal cells. Because of the urgent need for the use of marine antifouling coatings without biocides, this study investigates the antifouling behavior of thermo-responsive coatings against marine fouling, whereas responsive polymer coatings have not been used in this field. In other word, the purpose of this research is to examine the seawater temperature change as a natural driving force for marine fouling release. For this purpose, the PNIPAAm-responsive polymer was immobilized on glass substrates and its fouling release properties were investigated. The results show that, by adjusting the grafting density and the thickness of the polymeric layer, the coatings can effectively release almost all the attached amphora.

2. Experimental

2.1. Materials and methods

N-isopropyl acrylamide (NIPAAm) monomer, (3-aminopropyl)trimethoxysilane (APTMS), and 2-bromoisobutyryl bromide (BIBB) as ATRP initiator were purchased from Sigma-Aldrich. To purify the monomer and initiator, NIPAAm was crystallized from hexane, and BIBB was mixed with glacial acetic acid at 90 °C for 24 h and subsequently filtered and washed with diethyl ether and ethanol, respectively and then vacuum dried. All of the solvents were dried by molecular sieve at reflux conditions for 24 h and distilled under reduced pressure. The copper(I) bromide (CuBr) was purchased from Alfa Aesar. All of the other materials were prepared from Merck.

To graft the PNIPAAm on the surface of the glass substrate, the following steps were carried out. After washing with detergents, the float glass substrates were cleaned by piranha solution (H_2SO_4 to H_2O_2 ratio is 3:1) for 45 min at 110 °C and rinsed with deionized water. The substrates were dried with a nitrogen gun. Different levels of hydroxyl functionalization were induced by UV- O_3 exposure for 10, 30, and

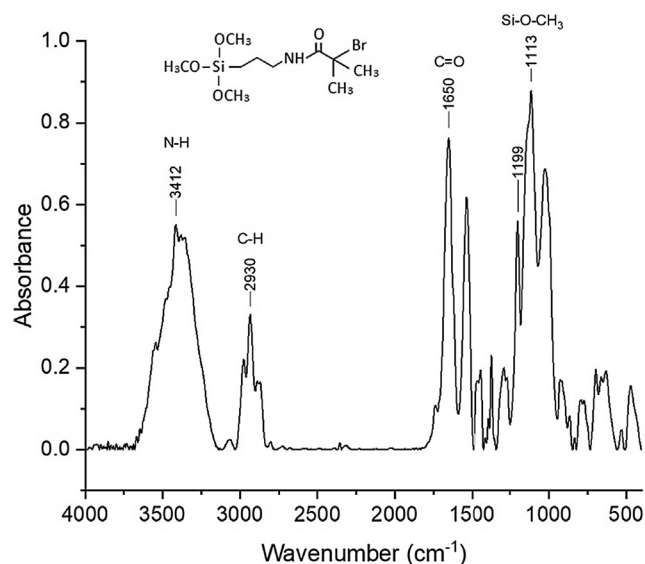


Fig. 1. FTIR spectrum of the synthesized initiator.

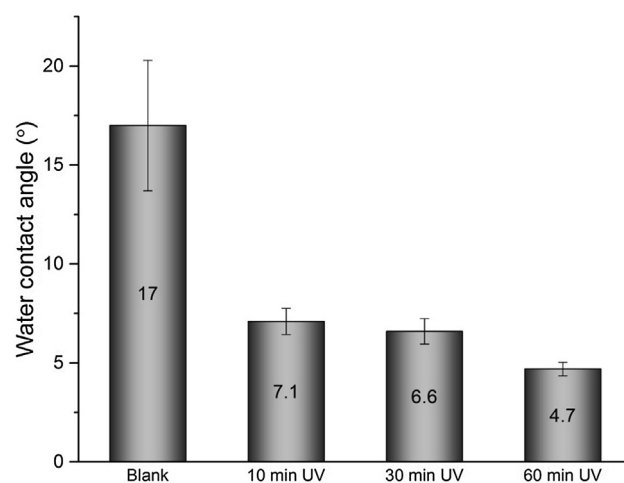


Fig. 2. WCA of glass substrates versus UV- O_3 exposure time.

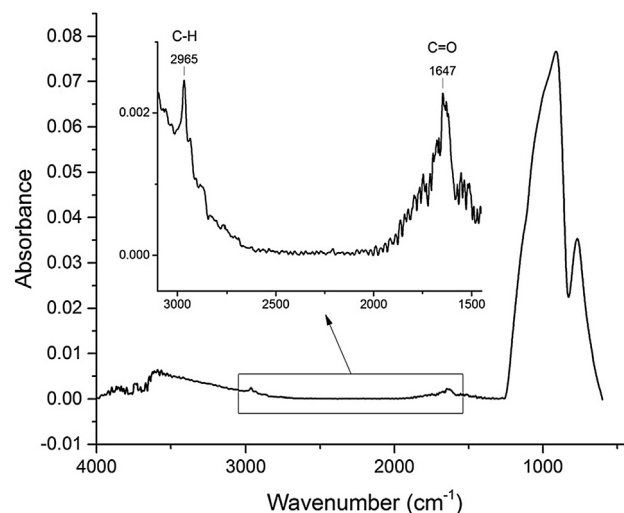


Fig. 3. ATR-FTIR spectrum of the PNIPAAm grafted glass substrate.

Table 1
Elemental analysis of the PNIPAAm grafted substrates.

Sample	Atomic %								
	C	N	O	Na	Mg	Al	Si	Ca	Sn
10 min UV- 4 h P	6.33	3.37	70.86	7.15	1.25	0.17	10.10	0.52	0.25
30 min UV- 4 h P	8.05	5.66	68.06	7.18	1.80	0.48	8.22	0.30	0.25
60 min UV- 4 h P	7.68	6.73	66.99	7.25	1.81	0.55	8.36	0.31	0.31
60 min UV- 2 h P	7.47	5.29	68.32	7.23	1.69	0.50	8.81	0.45	0.26
60 min UV- 6 h P	10.42	6.05	68.70	6.75	1.35	0.34	6.22	0.07	0.10

60 min. The BIBB was coupled to APTMS by dropwise adding 2.5 ml of BIBB to a solution containing 2.8 ml triethylamine, 2.9 ml APTMS, and 25 ml tetrahydrofuran (THF). The reaction was carried out under nitrogen atmosphere at 0 °C for one day. Precipitates were removed by filtration and centrifugation (12,000 rpm), and the product was obtained by solvent evaporation under vacuum.

The synthesized initiator was immobilized on the surface according to the following procedure. About 1 g (0.329 mmol) initiator was dissolved in 30 ml dried ethanol, and the freeze-pump-thaw cycle was performed for three times. The solution was transferred to a nitrogen glove box and was poured onto the substrates. The reaction carried out for 24 h. Finally, they were washed by ethanol.

NIPAAm (0.11 mmol) and N,N,N',N'',N''-pentamethyldiethylenetriamine (PMDTA) were dissolved in 20 ml deionized water and 20 ml isopropyl alcohol solution, and the solution was degassed with three freeze-pump-thaw cycles. The solution was mixed with CuBr (0.07 mmol) in the nitrogen glove box. The substrates were placed into the solution, and NIPAAm polymerization was carried out for 2, 4, and 6 h, and finally, they were washed with isopropyl alcohol. The presence of PNIPAAm on the glass substrates was confirmed by ATR-FTIR (Bruker, Tensor 27). The thickness of grafted thin films was investigated using SEM (Tescan model MV2300) of the cross-sectional area of samples broken in liquid nitrogen.

The adhesion of marine amphora to the substrates was investigated according to a standard method [43]. A suspension of 2.0×10^6 amphora/ml was poured on the modified glass substrates and was incubated in the dark at 20 °C for two hours. Some samples were incubated for a further 30 min at 37 °C to consider the high-temperature effect. The substrates were washed ten times with seawater to detach the free amphora. The temperature of the water was 20 and 37 °C for low and high-temperature experiments, respectively. The adhered amphora was fixed by using a 2% glutaraldehyde solution for 15 min. Finally, the substrates were washed by seawater, 50% seawater, and deionized water, respectively. The attached amphora number was measured by an optical microscope (Olympus BX41 equipped with Nikon camera) using a 40X objective lens. The number of replicates in each experiment was four times. Twenty images from different parts of a sample were recorded, and the final attached amphora number was the average of them.

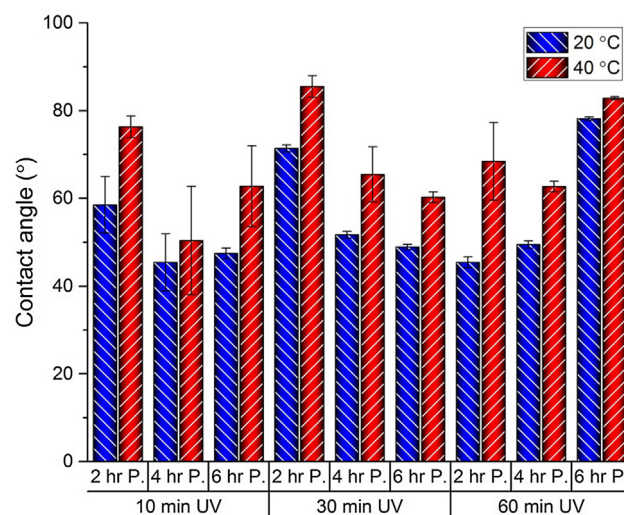


Fig. 5. WCA of PNIPAAm grafted substrates with different UV-O₃ irradiation and polymerization time at 20 and 40 °C.

3. Results and discussion

The synthesized initiator was characterized by the FTIR spectroscopy. The FTIR spectrum of the initiator shown in Fig. 1. The peaks corresponding to the N-H and C=O stretches were observed at 3412 and 1650 cm⁻¹, respectively. On the other hand, Si-O-CH₃ vibrations appeared at 1199 and 1113 cm⁻¹ wavenumbers, which indicated that the coupling reaction was performed successfully.

Different levels of hydroxyl functionalization of the glass substrates were induced using UV-O₃ irradiation. The extent of functionalization was revealed by WCA (Fig. 2). While the cleaned glass substrates were hydrophilic, by increasing the exposure time, the surface gets more hydrophilic properties. So that after 1 h of UV-O₃ irradiation the WCA reaches to about 4.7°.

ATR-FTIR spectroscopy was used to confirm the PNIPAAm grafting to the surface. In ATR-FTIR spectroscopy, the surface was characterized for about 2 μm depth of the sample, therefore for nanometric thin layers absorption peaks of the substrate were also observed in the spectrum and were stronger than those of the thin layer. ATR-FTIR of the PNIPAAm grafted substrate is shown in Fig. 3, and for better representation of the polymer thin layer absorptions, the result was zoomed for 1450–3100 cm⁻¹ and is shown as the figure inset. The C=O stretch vibration of the amide group and C-H stretch are at 1647 and 2965 cm⁻¹, respectively, which confirms the successful grafting of PNIPAAm to the surface. The sharp peak at 911 cm⁻¹ corresponds to the glass substrate.

The results of the energy-dispersive X-ray spectroscopy of the samples are shown in Table 1. Although the elements of the glass substrate were detected, by considering the C content of the samples, a

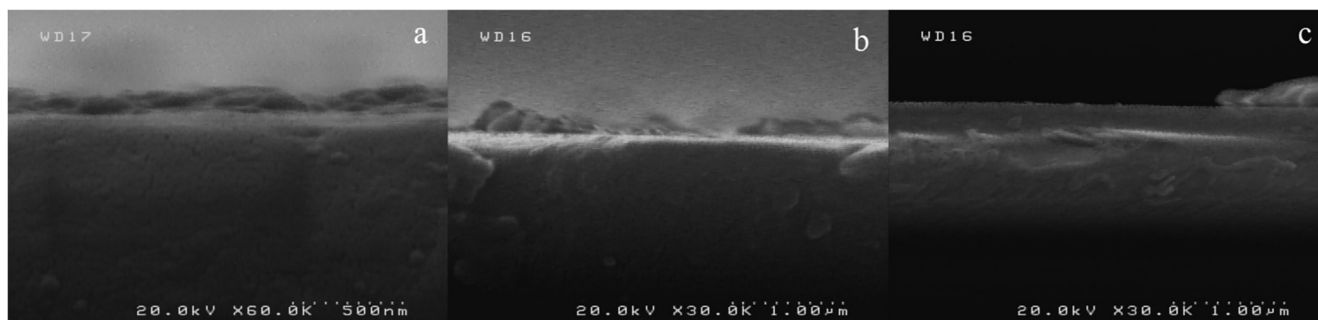


Fig. 4. SEM of the cross-section area of PNIPAAm grafted substrates with 60 min UV irradiation and (a) 2 h, (b) 4 h, and (c) 6 h polymerization time.

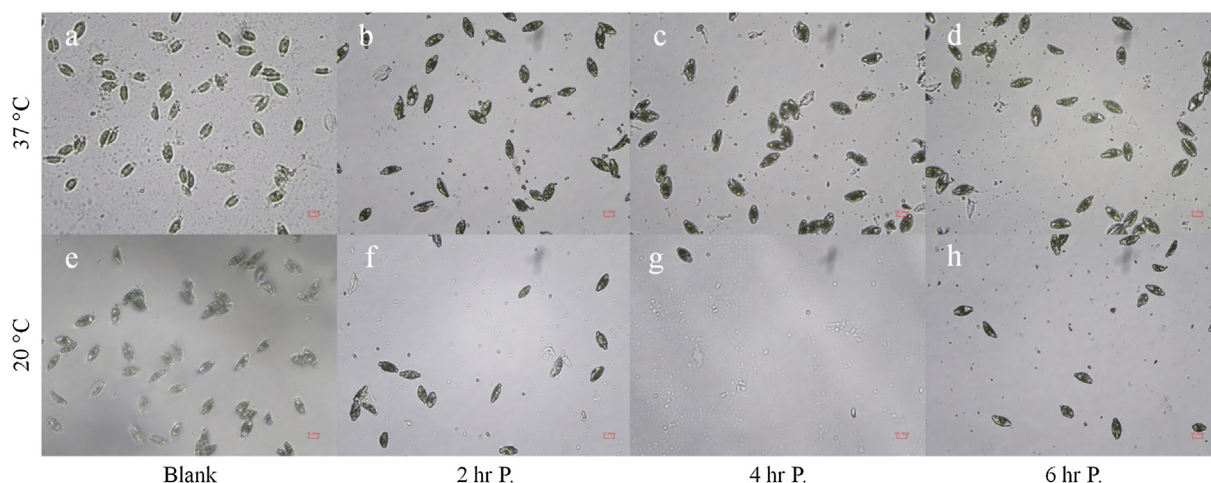


Fig. 6. Optical microscope images of the samples with 60 min UV-O₃ irradiation and polymerization time at 20 and 37 °C after adhesion of amphora, scale bars are 10 μm.

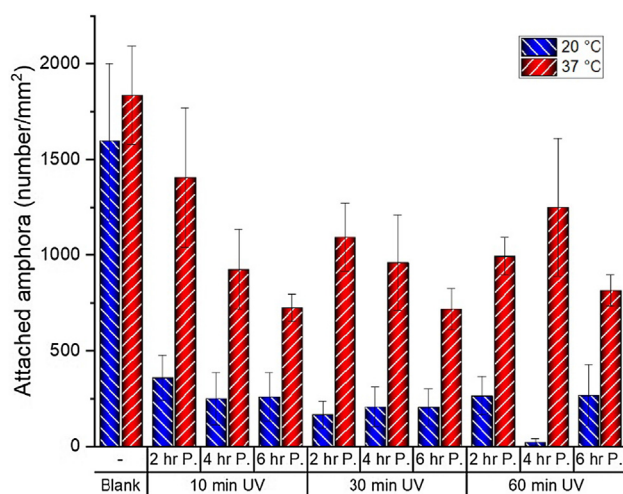


Fig. 7. The number of attached amphora on the substrates at different UV time, polymerization time, and temperature levels.

Table 2

The Summary of statistical analysis of the attached amphora using three-way ANOVA.

	DF	Mean Square	F-value	P-value
Polymerization time	2	1387108.86	45.87	0
UV time	2	271146.21	8.97	1.60×10^{-4}
Temperature	1	5.27×10^7	1743.48	0
Polymerization time $\times$ UV time	4	335674.04	11.10	1.78×10^{-8}
Polymerization time $\times$ Temperature	2	1475564.07	48.79	0
UV time $\times$ Temperature	2	111037.21	3.67	0.026
Polymerization time $\times$ UV time $\times$ Temperature	4	529093.31	17.50	4.49×10^{-13}
Model	17	3686674.25	121.91	0

qualitative investigation of the amount of polymeric layer is possible. Elemental analysis reveals that by increasing the UV-O₃ irradiation for 4 h polymerization time, the polymer content increased, which was an indication of the higher grafting density on the substrates. Also, by increasing the polymerization time from 2 to 6 h, the C content rises from 7.47% to 10.42% as a result of a thicker polymer layer. Therefore, by manipulating the UV-O₃ irradiation and polymerization time, the grafting density and the thickness of the polymeric layer were adjusted.

The results of the SEM of the PNIPAAm thin-film cross-section are

shown in Fig. 4. Although the thickness of the film in each sample was variable, by increasing the polymerization time and subsequent longer chain length of brushes, the thickness increased nearly linearly, which indicated that the grafted polymer chains were in the brush-like regime.

The thermo-responsiveness of the grafted layers was investigated using WCA analysis. The WCA of the samples was measured at 20 °C and 40 °C, below and above the LCST of NIPAAm (32 °C), and the results were presented in Fig. 5. For all the samples, by increasing temperature, WCA increased, and the changes ranged from 5 to 23°. As a general trend, for higher polymerization times, the WCA difference between the high and low temperatures decreased, indicating a thicker layer is not essentially better than a thinner one. So, it would be an optimum thickness in which maximum change in WCA occurred. On the other hand, at a constant polymerization time (2 hr), by increasing the UV-O₃ irradiation from 10 to 60 min, which corresponded to a higher grafting density, the WCA difference between the high and low temperature increased while for 6 h polymerization time, it decreased. Therefore, while a dense grafting of the short chains made the surface more sensible, longer chains had a reverse effect.

Adhesion of microorganisms to the samples was investigated by a standard procedure using amphora as the marine fouling, and the results were presented in Figs. 6 and 7. For cleaned glass substrates without any grafted polymer (blank), the number of attached amphora was very high, and temperature reduction from 37 °C to 20 °C slightly reduced them. By incorporating the PNIPAAm, the number of attached amphora at high temperature decreased. While the temperature lowering had a little effect on the blank sample, for PNIPAAm grafted ones, the release of the attached amphora significantly enhanced. So that for the sample with 60 min UV-O₃ irradiation and 4 h polymerization, nearly all the attached amphora released. The WCA difference between the high and low temperatures of this sample was about 14°, which was not the highest one. It can conclude that not only the surface should have a reasonable thermo-responsive nature, but also a suitable grafting density was needed.

Statistical analysis of the results was performed using a 3-way ANOVA method using OriginLab software. For this purpose, three independent factors, including polymerization time, UV irradiation time, and temperature, were considered at 3, 3, and 2 levels, respectively, and the response was the number of the attached amphora.

The experimental design was based on full factorial, which takes into account all main and interaction effects. The results of the statistical analysis are shown in Table 2. All the factors and interactions have a P-value less than 0.05 and an F-value higher than one, indicating that all of them are significant. F-values show that all the effects that comprise the UV irradiation time are less important than the others.

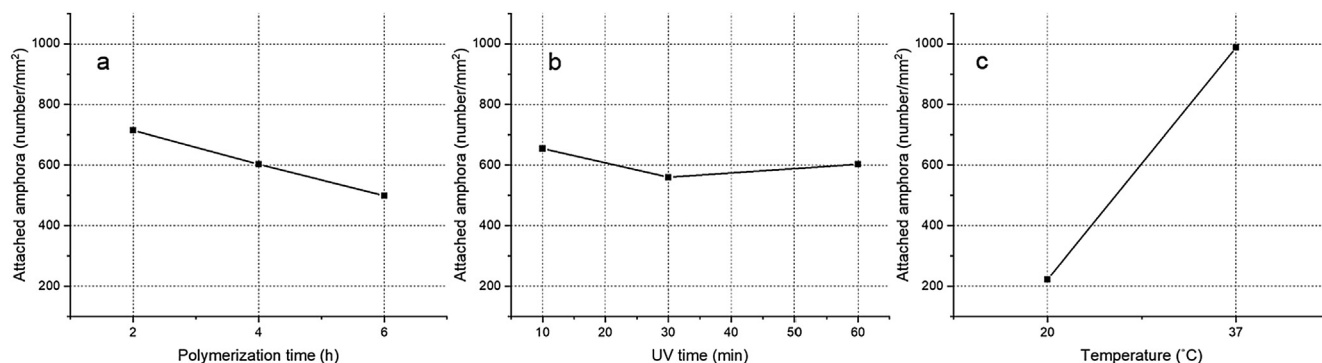


Fig. 8. The main effects of independent factors, polymerization time (a), UV time (b), and temperature (c) on the attached amphora number.

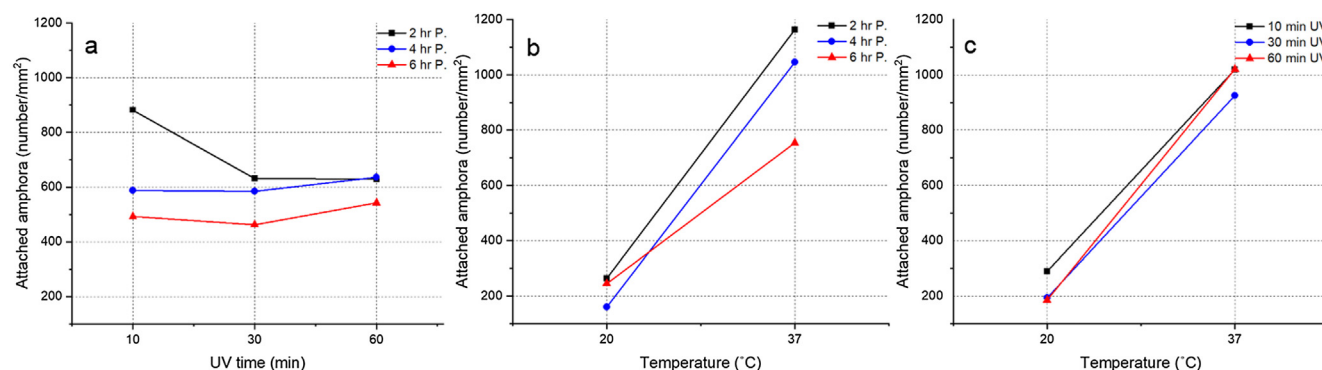


Fig. 9. The interaction effects of independent factors, polymerization time × UV time (a), polymerization time × temperature (b), and UV time × temperature (c) on the attached amphora number.

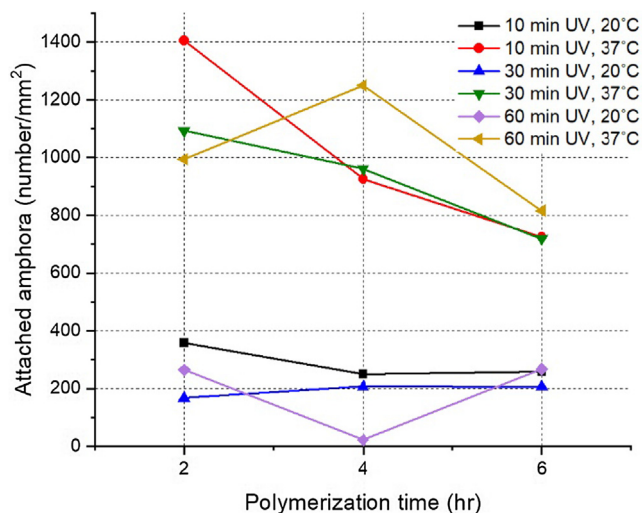


Fig. 10. The three-way interaction effects of independent factors, polymerization time × UV time × temperature on the attached amphora number.

The main effects of independent factors are shown in Fig. 8. By increasing the polymerization time from 2 to 6 h, the attached amphora on the surface decreased, suggesting that a thicker film releases the amphora more efficiently. While the effect of UV time was significant, in comparison to other factors has a weaker effect. Among these factors, the temperature is the most effective one so that with the decrease in temperature from 37 to 20 °C, the amount of attached amphora decreases sharply. This significant evidence proves that thermo-responsive polymers have a profound ability to release the attached amphora.

One of the benefits of full factorial design is the ability to investigate

the interaction effect of independent factors. The interaction effects of different combinations of factors are shown in Fig. 9. By increasing the UV irradiation time from 10 to 30 min at 2 h polymerization time, the release of amphora increased by 30% while subsequent increasing had a minor effect. So, the polymer thin film corresponding to 10 min UV irradiation and 2 h polymerization did not cover the entire surface while 30 min UV time was sufficient. At high temperature, increasing the polymerization time decreases the attached amphora while at low temperature, 4 h polymerization was more efficient. Generally, the effect of UV irradiation time was weak, whereas lowering the temperature had a strong effect on releasing the attached amphora.

The three-way interaction effects of independent factors (Fig. 10) show that the attached amphora sharply reduced due to lowering the temperature. The three-way interaction effect at 4 h polymerization time, 60 min UV time, and low-temperature levels was very effective. As shown in Fig. 7, the combination of main effects and interactions, especially three-way interaction at these levels made this sample very efficient and completely removed the attached amphora.

4. Conclusions

The idea of using water temperature change as a natural energy source to release the marine fouling was examined. The results showed that the PNIPAAm thermo-responsive coatings successfully released the attached amphora from the glass substrates. Controlled by ATRP time, the polymer chain length has a significant effect on the fouling release properties of the coating. Since the LCST of PNIPAAm is higher than the mean temperature of seawaters, it is necessary to adjust the LCST of the coating to about 10–15 °C. This can be achieved by copolymerization of NIPAAm with other monomers or using other thermo-responsive polymers. Therefore, thermo-responsive polymers can release marine microorganisms, and they can be suitable and environmentally friendly alternatives for toxic antifouling coatings.

CRediT authorship contribution statement

M. Rahimi: Visualization, Formal analysis, Validation, Investigation. **M. Nasiri:** Conceptualization, Methodology, Formal analysis, Resources, Writing - original draft, Writing - review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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